Fedosov quantisation

In this section, we introduce the 'simple geometrical construction' of a star-product on any symplectic manifold due to Fedosov [1, 2]. The idea is the following: given that on symplectic manifolds, one has the Darboux theorem, which tells us that for any point there always a neighbourhood containing it that is symplectomorphic to the flat symplectic space $(\mathbb{R}^{2n}, \omega_{\text{can.}})$, then one can find *locally* a star-product, in the guise of the Moyal–Weyl product. In other words, the Moyal–Weyl product gives a local expression for the star-product on a symplectic manifold. The question therefore boils down to whether it is possible to *patch* together all of these local solutions to define a star-product *globally*.

This was first shown by De Wilde and Lecomte [3] by cohomological arguments: they proved the existence of star-products on symplectic manifolds, but did not give a general formula for it. Fedosov's insight was to prove this theorem in a constructive manner, by explicitly working out 'how to patch' the star-product formulae given by the Moyal–Weyl product in each Darboux patch.

(*i*) First, one construct a bundle, whose fibre above each point is isomorphic to the algebra of polynomial functions on $\mathbb{R}^{2n}$, the 'flat model' for symplectic manifolds;

(*ii*) Second, one endows this bundle with a flat connection, allowing one to relate functions on M to sections of it.

(*iii*) Third, one can finally define a star-product on M globally by sending functions to their associated sections, using the Moyal–Weyl product fibrewise to get another section, and projecting the result back to a function on the base.

The Weyl bundle. Given a symplectic vector space (V, ω), that is, a vector space V together with a bilinear, antisymmetric and non-degenerate form $\omega : V \times V \to \mathbb{R}$, one can define the associated Weyl algebra as

$$\mathcal{W}_V := \big(S(V^*)[\![\hbar]\!], *\big)\,, \tag{1}$$

where $*$ denotes the Moyal–Weyl star-product. Indeed, choosing a basis of V, say $\mathtt{t}_a$ with $a = 1, \dots, 2n$, is equivalent to choosing an isomorphism $V \cong \mathbb{R}^{2n}$, which is the same as a choice of coordinates on V (seen as a flat manifold), say y^a. The symmetric algebra $S(V^*)$ is then identified with the algebra of polynomial functions on $\mathbb{R}^{2n}$, for which the Moyal–Weyl product defines a deformation quantization. The resulting associative but non-commutative algebra is called the *Weyl algebra*. With the aforementioned choice of coordinates,

$$\mathcal{W}_V \cong \mathcal{W}_{\mathbb{R}^{2n}} \cong \mathbb{R}[y^a][\![\hbar]\!]\,, \tag{2}$$

and the associative Moyal–Weyl product on its generators y^a reads

$$y^a * y^b = y^a y^b + \tfrac{\hbar}{2}\omega^{ab}\,, \tag{3}$$

with ω^{ab} the component of the inverse of the symplectic form ω in the basis dual to $\mathtt{t}_a$.[1]

For instance, on any symplectic manifold, the tangent space (T_pM, ω_p) at each point $p \in M$ is a symplectic vector space. This enables one to define one of the crucial ingredient in Fedosov's construction, which is the Weyl bundle.

Definition 1 [**Weyl bundle**]**.** Let (M, ω) be a symplectic manifold. The Weyl bundle over M is defined as

$$\mathcal{W} := S(T^*M)[\![\hbar]\!] \longrightarrow\!\!\!\!\!\rightarrow M \tag{4}$$

is a bundle of associative algebras over M, whose fiber over a point $p \in M$ is the Weyl algebra $\mathcal{W}_{T_pM}$ associated with the tangent space at this point.

[1] Here we are using the convention $\omega^{ac}\omega_{bc} = \delta^a_b$ for the inverse.

Concretely, sections of the Weyl bundle take the form

$$\Gamma(\mathcal{W}) \ni F(x;y) = \sum_{k,n\geq 0} \hbar^k F^{(k)}_{a_1\dots a_n}(x)\, y^{a_1}\dots y^{a_n}\,, \tag{5}$$

with y^a the generators of the Weyl algebra, as denoted before, in the fiber above the point $x \in M$, which can also be considered as a basis of the cotangent space at this point (i.e. it defines a coframe), and the components $F^{(k)}_{a_1\dots a_n}(x)$ are those of a totally symmetric covariant tensor of rank n. Put differently, these sections are formal power series in $\hbar$, with coefficients in totally symmetric covariant tensors on M.

The space of forms valued in the Weyl bundle, $\Omega(M,\mathcal{W})$, is a complex with differential, which locally takes the form

$$\delta := \mathrm{d}x^a \tfrac{\partial}{\partial y^a}\,, \tag{6}$$

where x^a are coordinates on the symplectic manifold (M,ω), and y^a a basis of the cotangent space over the point x. This differential is inner, in the sense that

$$\delta = \tfrac{1}{\hbar}[\omega_{ab}\, y^a\, \mathrm{d}x^b, -]_*\,, \tag{7}$$

and a simple computation leads to

$$\delta^2 = \tfrac{1}{\hbar}\,[\omega, -]_* = 0\,, \tag{8}$$

since the symplectic 2-form ω takes values in the center of the Weyl algebra.

Another important operator on forms valued in the Weyl bundle is the following

$$h = \tfrac{1}{N}\, y^a \imath_{\partial_a}\,, \tag{9}$$

where $\imath_{\partial_a}$ denotes the interior product with the basis vector $\frac{\partial}{\partial x^a}$, and N is an operator that returns the sum of the form degree and the degree in y of the element on which its acts. A direct computation shows that

$$\delta\, h + h\, \delta = \begin{cases} 0 & \text{on } \Omega^0(M,\mathcal{W}^0)\,, \\ \mathbf{1} & \text{otherwise}\,, \end{cases} \tag{10}$$

i.e. h defines a *contracting homotopy* for the differential δ. Consequently, the cohomology of $\Omega(M,\mathcal{W})$ is almost empty, expect in for degree 0.

Lemma 1. *The complex* $\big(\Omega(M,\mathcal{W}),\delta\big)$ *is acyclic, i.e. its cohomology is concentrated in degree* 0*, and* $H^0(\delta) \cong \mathcal{C}^\infty(M)$*.*

Proof. This is a simple consequence of the contracting homotopy h: indeed, if $\alpha \in \Omega^p(M,\mathcal{W})$ with $p > 0$, then

$$\delta\alpha = 0 \qquad \Rightarrow \qquad 0 = -\delta\, h\, \alpha + \alpha\,, \tag{11}$$

upon using (10), which implies that any cocycle α is also a coboundary (i.e. in the image of δ), and hence $H^p(\delta) = \{0\}$ for any strictly integer $p > 0$. For $\phi \in \Omega^0(M,\mathcal{W})$, the cocycle condition

$$\delta\phi = \mathrm{d}x^a \tfrac{\partial}{\partial y^a}\phi(x,y) = 0 \qquad \Rightarrow \qquad \phi \in \Omega^0(M,\mathcal{W}^0)\,, \tag{12}$$

implies that ϕ is y-independent, so that $\phi \in \Omega^0(M,\mathcal{W}^0) \cong \mathcal{C}^\infty(M)$, hence the lemma. ∎

In other words, $\big(\Omega(M,\mathcal{W}),\delta\big)$ defines a resolution of $\mathcal{C}^\infty(M)$ — the latter is recovered as the cohomology of the former. Another way to summarise this result is that there exists an Hodge decomposition

$$\Omega(M,\mathcal{W}) \cong \mathrm{Im}(\delta) \oplus \mathrm{Im}(h) \oplus \mathrm{Ker}(\{h,\delta\})\,, \tag{13}$$

which is to say that any $\alpha \in \Omega(M,\mathcal{W})$ can be written as

$$\alpha = \alpha|_{y=0} + \delta h(\alpha) + h\delta(\alpha)\,. \tag{14}$$

Fedosov connections. The central object of Fedosov's construction is an 'Abelian connection', to use his terminology, which can be thought of as a prolongation of a linear, torsion-free, symplectic connection, into a connection on the Weyl bundle whose curvature is central. Before spelling out this construction, let us first recall the definition of a symplectic connection.

Definition 2 [**Symplectic connection**]**.** An affine connection ∇ on a symplectic manifold (M, ω) is called symplectic if it is

(i) Torsion-free, i.e.
$$\nabla_X Y - \nabla_Y X - [X, Y] = 0\,, \tag{15}$$
for any $X, Y \in \Gamma(TM)$;

(ii) Compatible with ω, i.e. $\nabla\omega = 0$, or more explicitly
$$X[\omega(Y, Z)] = \omega(\nabla_X Y, Z) + \omega(Y, \nabla_X Z)\,, \tag{16}$$
for any $X, Y, Z \in \Gamma(TM)$.

The above two conditions read, in components,
$$\Gamma^c_{[ab]} = 0 = \partial_a \omega_{bc} + 2\,\Gamma^d_{a[b}\,\omega_{c]d}\,, \tag{17}$$
where Γ^c_{ab} are the coefficients of the connection ∇.

Lemma 2. *Symplectic connections exist on any symplectic manifold, but are not unique. More precisely, the set of symplectic connections compatible with a given symplectic 2-form ω is endowed with a structure of an affine space modelled on the vector space of symmetric rank-3 covariant tensors.*

Proof. Let ∇ be a torsionfree connection on a symplectic manifold with symplectic 2-form ω and define the associated 'symplecticity' tensor $C_{a|bc} := \partial_a \omega_{bc} + 2\,\Gamma^d_{a[b}\,\omega_{c]d}$ as the obstruction for ∇ to be compatible with ω. The latter tensor has two irreducible components

- The hook part, $C^{\ydiagram{2,1}}_{a,bc} := C_{a|bc} - C_{[b|c]a} = \partial_a \omega_{bc} - \partial_{[b}\omega_{c]a} + 3\,\Gamma^d_{a[b}\,\omega_{c]d}$;
- The antisymmetric part, $C^{\ydiagram{1,1,1}}_{abc} := C_{[a|bc]} = \partial_{[a}\omega_{bc]} = 0$, which vanishes identically by virtue of the fact that ω is closed.[2]

Similarly, the components of ∇ with all indices down, $\Gamma_{ab|c} := \Gamma^d_{ab}\,\omega_{cd}$, have two irreducible pieces

- A symmetric part, $\Gamma^{\ydiagram{3}}_{abc} := \Gamma_{(ab|c)}$;
- A hook part, $\Gamma^{\ydiagram{2,1}}_{ab,c} := \Gamma_{ab|c} - \Gamma_{c(a|b)}$;

in terms of which the unprojected coefficients are given by $\Gamma_{ab|c} = \Gamma^{\ydiagram{3}}_{abc} + \frac{2}{3}\,\Gamma^{\ydiagram{2,1}}_{ab,c}$. Plugging this decomposition back into the 'symplecticity' condition, one finds
$$C^{\ydiagram{2,1}}_{a,bc} = 0 \qquad \Longleftrightarrow \qquad \partial_a \omega_{bc} - \partial_{[b}\omega_{c]a} + 2\,\Gamma^{\ydiagram{2,1}}_{a[b,c]} = 0\,, \tag{18}$$
which, upon taking the symetric part in (ab) and using the hook symmetry, leads to
$$\Gamma^{\ydiagram{2,1}}_{ab,c} = -\partial_{(a}\omega_{b)c}\,. \tag{19}$$
Hence, the hook part of the 'symplecticity' condition completely fixes the hook part of the connection in terms of ω. Furthermore, the totally skewymmetric part of the 'symplecticity' condition

[2] Note that this is a necessary condition for a 2-form ω to be preserved by a torsionless connection: a direct computation yields
$$\nabla_{[a}\omega_{bc]} = \partial_{[a}\omega_{bc]} + T^d_{[ab}\,\omega_{c]d}\,,$$
or equivalently,
$$\tfrac{1}{3}\,(\mathrm{d}\omega)(X, Y, Z) = (\nabla_X \omega)(Y, Z) + \omega(T(X, Y), Z) + \text{cyclic perm.}\,, \qquad X, Y, Z \in \Gamma(TM)\,,$$
where T is the torsion of ∇, which shows that if ω is preserved by a torsionless connection ∇, then ω is closed.

is automatically satisfied thanks to the fact that ω is closed. Hence, plugging back the expression of $\Gamma^{\boxplus}$ into the expression for the unprojected coefficients completely solves the 'symplecticity' condition. The resulting most general form of a symplectic connection reads

$$\Gamma^c_{ab} = -\tfrac{2}{3}\,\omega^{cd}\partial_{(a}\omega_{b)d} + \omega^{cd}\Sigma_{abd}$$

where $\Sigma \in \Gamma(S^3T^*M)$ is a totally symmetric covariant rank-3 tensor encoding the arbitrariness into ∇. ■

For more details on symplectic connections, see e.g. the extensive and pedagogical review [4]. Having introduced symplectic connections, we can now state one of Fedosov's main theorem, which is can any such connection can be completed into a connection on the Weyl bundle whose curvature is *central*, i.e. proportional to the unit of the Weyl algebra.

Let us first note that any symplectic connection naturally acts on sections of the Weyl bundle,[3] the covariant derivative locally reads

$$\nabla = \mathrm{d} + \tfrac{1}{\hbar}[\Gamma, -]_*\,, \qquad \text{with} \qquad \Gamma := -\tfrac{1}{2}\,\Gamma^d_{(ab}\omega_{c)d}\,y^a y^b \mathrm{d}x^c\,, \tag{20}$$

and its curvature is given by

$$\nabla^2 = \tfrac{1}{\hbar}\,[R, -]_*\,, \qquad \text{with} \qquad R = -\tfrac{1}{4}\mathrm{d}x^a\mathrm{d}x^b\,R_{ab}{}^{\bullet}{}_{c}\,\omega_{d\bullet}\,y^c y^d\,, \tag{21}$$

where

$$R_{ab}{}^c{}_d = 2\big(\partial_{[a}\Gamma^c_{b]d} + \Gamma^c_{[a|\times}\Gamma^{\times}_{b]d}\big)\,, \tag{22}$$

are the components of the curvature tensor of ∇. Notice also that the graded-commutator of this covariant derivative with the operator δ previously introduce vanishes,

$$[\delta, \nabla] \equiv \delta\nabla + \nabla\delta = 0\,, \tag{23}$$

as a consequence of the fact that ∇ is torsion-free. The two operators δ and ∇ therefore define a degree 1 derivation on the space $\Omega(M, \mathcal{W})$ of forms valued in the Weyl bundle, whose square encodes the torsion and curvature of ∇,

$$(-\delta + \nabla)^2 = \tfrac{1}{\hbar}[R, -]_*\,, \tag{24}$$

where the minus sign before δ is introduced for later convenience. While one can always choose a torsionless connection, as we did so far, flat connection do not always exist.

One can think of the torsion and curvature of ∇ as the obstruction for the combination $-\delta+\nabla$ to be a *differential* on $\Omega(M, \mathcal{W})$. These two obstructions can be separated by introducing a grading, namely by assigning to y^a and $\hbar$ degree 1 and 2 respectively,

$$\deg(\hbar) = 2\,, \qquad \deg(y^a) = 1\,, \tag{25}$$

so that

$$\deg(*) = 0\,, \qquad \deg(\delta) = -1\,, \qquad \deg(\nabla) = 0\,, \tag{26}$$

as operators acting on forms valued in the Weyl bundle. We will refer to this grading as the Fedosov degree from now on. The sum $-\delta + \nabla$ is therefore an inhomogeneous derivation, with a piece in degree -1 and one in degree 0, and hence its square has components in degrees -2, -1 and 0, respectively corresponding to the square of δ, the torsion of ∇ and its curvature.

The idea of the following theorem is to complete $-\delta + \nabla$ with the adjoint action of a certain 1-form valued in the Weyl bundle defining an operator with components of degree ≥ 1, such that the it squares to zero.

[3] Recall that the symplectic algebra $\mathfrak{sp}(2n, \mathbb{R})$ is embedded in the Weyl algebra $\mathcal{W}_{\mathbb{R}^{2n}}$, as quadratic elements, and acts on $\mathcal{W}_{\mathbb{R}^{2n}}$ via the commutator. This action integrates to give a representation of the Lie group $Sp(2n, \mathbb{R})$ on $\mathcal{W}_{\mathbb{R}^{2n}}$, which gives us another way of defining the Weyl bundle, namely as the associated bundle

$$\mathcal{W} = F_{Sp}(M) \times_{Sp(2n,\mathbb{R})} \mathcal{W}_{\mathbb{R}^{2n}}\,,$$

where $F_{Sp}(M)$ denotes the $Sp(2n, \mathbb{R})$-principal bundle of symplectic frames.

Theorem 3 (Fedosov [1]). *Given any symplectic connection ∇, and formal power series*

$$\omega_\hbar = \omega + \sum_{k=1}^{\infty} \hbar^k \, \omega_k \,, \qquad \omega_k \in \Omega^2(M) \,, \quad \forall k \geq 1 \,, \tag{27}$$

of closed 2-forms, there exists a unique Weyl algebra-valued 1-form $\gamma \in \Omega^1(M, \mathcal{W}^{\geq 3})$ obeying $h\gamma = 0$ and such that

$$\mathcal{D} := -\delta + \nabla + \tfrac{1}{\hbar}[\gamma, -]_* \equiv \mathrm{d} + \tfrac{1}{\hbar}[A, -]_* \,, \tag{28}$$

defines a connection on the Weyl bundle whose curvature is given by

$$\mathrm{d}A + \tfrac{1}{2\hbar}[A, A]_* = -\omega_\hbar \,, \tag{29}$$

and hence lies in the center of the Weyl algebra.

Proof. Let us assume that $\gamma \in \Omega^1(M, \mathcal{W}^{\geq 3})$ such that

$$\mathcal{D} := -\delta + \nabla + \tfrac{1}{\hbar}[\gamma, -]_* \,, \tag{30}$$

is flat, exists. The flatness of $\mathcal{D}$ then reads

$$\mathcal{D}^2 = 0 = \tfrac{1}{\hbar}\left[\omega - \delta\gamma + R + \nabla\gamma + \tfrac{1}{2\hbar}\left[\gamma, \gamma\right]_*, -\right]_* \,, \tag{31}$$

which amounts to the requirement that the 2-form appearing inside the $*$-commutator on the right hand side of the above equation takes values in the center of the Weyl bundle. In other words, there exists 2-forms $\omega_k \in \Omega^2(M)$ for $k \in \mathbb{N}^\times$ such that

$$\omega - \delta\gamma + R + \nabla\gamma + \tfrac{1}{2\hbar}\left[\gamma, \gamma\right]_* = -\sum_{k \geq 1} \hbar^k \, \omega_k \,. \tag{32}$$

On top of that, the Bianchi identity imposes that these 2-forms are all closed, $\mathrm{d}\omega_k = 0$ for all $k \geq 1$. Expanding this equation according to the Fedosov degree,

$$\delta\gamma_{(n+1)} = R\,\delta_{n,3} + \nabla\gamma_{(n)} + \tfrac{1}{2\hbar}\sum_{k=3}^{n-1}[\gamma_{(k)}, \gamma_{(n+2-k)}]_* + (\omega_\hbar)_{(n)} =: \beta_{(n)} \,, \tag{33}$$

one can notice that the right hand side only involves $\gamma_{(k)}$ with $k \leq n$. Assuming that $\gamma_{(k)}$ satisfying $\delta\gamma_{(k)} = \beta_{(k-1)}$ exists for $k \leq n$, the existence of $\gamma_{(n+1)}$ verifying the above equation is a cohomological problem: a direct computation shows that $\beta_{(n)}$ are cocycles themselves, so that we have to solve the equations

$$\delta\gamma_{(n+1)} = \beta_{(n)} \,, \qquad \delta\beta_{(n)} = 0 \,, \tag{34}$$

i.e. find a cochain making $\beta_{(n)}$ exact. The existence of $\gamma_{(n+1)}$ follows from the fact that $\Omega(M, \mathcal{W})$ is acyclic. What is more, we can find an explicit solution for γ using the contracting homotopy h, namely

$$\gamma_{(n+1)} = h\big(R\,\delta_{n,3} + \nabla\gamma_{(n)} + \tfrac{1}{2\hbar}\sum_{k=3}^{n-1}[\gamma_{(k)}, \gamma_{(n+2-k)}]_* + (\omega_\hbar)_{(n)}\big) \,, \tag{35}$$

upon imposing $h\gamma = 0$, which is consistent since $h^2 = 0$. ∎

Next, we can use the flat connection to define a *lift* of function on the base, symplectic, manifold (M, ω), to *covariantly constant* sections of the Weyl bundle.

Theorem 4 (Fedosov [1]). *The algebra of functions $\mathcal{C}^\infty(M)$ on a symplectic manifold is isomorphic, as a vector space, to the space of covariantly constant sections of the Weyl bundle, i.e. $\mathcal{C}^\infty(M) \cong H^0(\mathcal{D})$ where $\mathcal{D}$ is a Fedosov connection.*

Proof. This amounts to showing that, given $f \in \mathcal{C}^\infty(M)$, one can find a unique section $F \in \Omega^0(M, \mathcal{W})$ such that $\mathcal{D}F = 0$ and $F|_{y=0} = f$. The condition $\mathcal{D}F = 0$ can be solved iteratively, like in the proof of the previous theorem: expanding it according to the Fedosov degree,

$$\delta F_{(n+1)} = \nabla F_{(n)} + \tfrac{1}{\hbar}\sum_{k=0}^{n-1}[A_{(n+2-k)}, F_{(k)}]_* \,, \tag{36}$$

we see that one can solve for $F_{(n+1)}$ in terms of $\{F_{(k)}\}_{k\leq n}$ upon using the contracting homotopy. Explicitly,

$$F_{(n+1)} = h\big(\nabla F_{(n)} + \tfrac{1}{\hbar}\sum_{k=0}^{n-1}[A_{(n+2-k)}, F_{(k)}]_*\big)\,, \tag{37}$$

since $h(F) = 0$ (F being a zero-form). ■

The first few orders of the lift of $f \in \mathcal{C}^\infty(M)$ read

$$F = f + y^a\,\nabla_a f + \tfrac{1}{2}\,y^a y^b\,\nabla_a\nabla_b f + \tfrac{1}{6}\,y^a y^b y^c\,\big(\nabla_a\nabla_b\nabla_c f + \tfrac{1}{4}\,R_{a\bullet}{}^{\times}{}_{b}\,\omega_{c\times}\,\omega^{\bullet d}\,\nabla_d f\big) + \mathcal{O}(y^4)\,. \tag{38}$$

We can now define a star-product on (M,ω) via

$$f \star_\nabla g := (F * G)|_{y=0}\,, \tag{39}$$

where F and G are the lifts of f and g respectively. Its associativity follows directly from that of the Moyal–Weyl product.

One may wonder: What is the dependence of this construction on the choice of symplectic connection? First, note that the pair (A,F) where $A \in \Omega^1(M,\mathcal{W})$ is such that $\mathcal{D} = \mathrm{d} + \frac{1}{\hbar}\,[A,-]_*$ is a Fedosov connection and F the lift of a function $f \in \mathcal{C}^\infty(M)$ are solutions of the system of equations

$$\mathrm{d}A + \tfrac{1}{2\hbar}\,[A,A]_* = \omega_\hbar\,, \qquad \mathrm{d}F + \tfrac{1}{\hbar}\,[A,F]_* = 0\,, \tag{40}$$

i.e. an 'almost flatness' condition, fixing the curvature of A to be central, and a covariant constancy condition for the section F. As a consequence, these equations are invariant under the usual gauge transformations

$$A \to g_\xi^{-1} * (\hbar\mathrm{d} + A) * g_\xi\,, \qquad F \to g_\xi^{-1} * F * g_\xi\,, \qquad \text{with} \qquad g_\xi := \exp_*(\tfrac{1}{\hbar}\,\xi)\,, \tag{41}$$

for any $\xi \in \Omega^0(M,\mathcal{W})$. Such elements obviously form a group, that we will refer to as the Fedosov gauge group.

Theorem 5. *Given any two symplectic connections ∇ and $\tilde\nabla$, and their associated Fedosov connections $\mathcal{D}$ and $\tilde{\mathcal{D}}$, there exists a gauge parameter $\xi \in \Omega^0(M,\mathcal{W})$ such that A and $\tilde A$ are related by the gauge transformation it generates.*

Proof. Let us try to solve

$$\tilde A = g^{-1} * (\hbar\mathrm{d} + A) * g\,, \tag{42}$$

for $g \in \Omega^0(M,\mathcal{W})$. This equation can be re-written as

$$\delta g = \hbar\nabla g + [\gamma, g]_* + g * (A - \tilde A)\,, \tag{43}$$

Applying the contracting homotopy h, and expanding the equation according to the Fedosov degree, yields

$$g_{(n)} = h\Big(\hbar\,\nabla g_{(n-1)} + \sum_{k=3}^{n+1}[\gamma_{(k)}, g_{(n+1-k)}]_* - \sum_{k=2}^{n+2} g_{(n+1-k)} * (A_{(k)} - \tilde A_{(k)})\Big)\,, \tag{44}$$

which, again, allows us to solve for g iteratively, since only $g_{(k)}$ with $k < n$ appear on the right hand side of this equation. ■

References

[1] B. V. Fedosov, "A simple geometrical construction of deformation quantization," *J. Diff. Geom.* **40** (1994) no. 2, 213–238.

[2] B. V. Fedosov, *Deformation quantization and index theory.* Akademie Verlag, 1996.

[3] M. De Wilde and P. B. Lecomte, "Existence of star-products and of formal deformations of the Poisson Lie algebra of arbitrary symplectic manifolds," *Letters in Mathematical Physics* **7** (1983) no. 6, 487–496.

[4] P. Bieliavsky, M. Cahen, S. Gutt, J. Rawnsley, and L. Schwachhofer, "Symplectic connections," *Int. J. Geom. Methods Mod. Phys.* **3** (2006) no. 03, 375–420, arXiv:math/0511194 [math.SG].